

Tutorial – 05

Name : Ranudi Gayathmie Kariyapperuma
Index : 29897
Batch No : 23.1
Degree Program Name : Technology Management

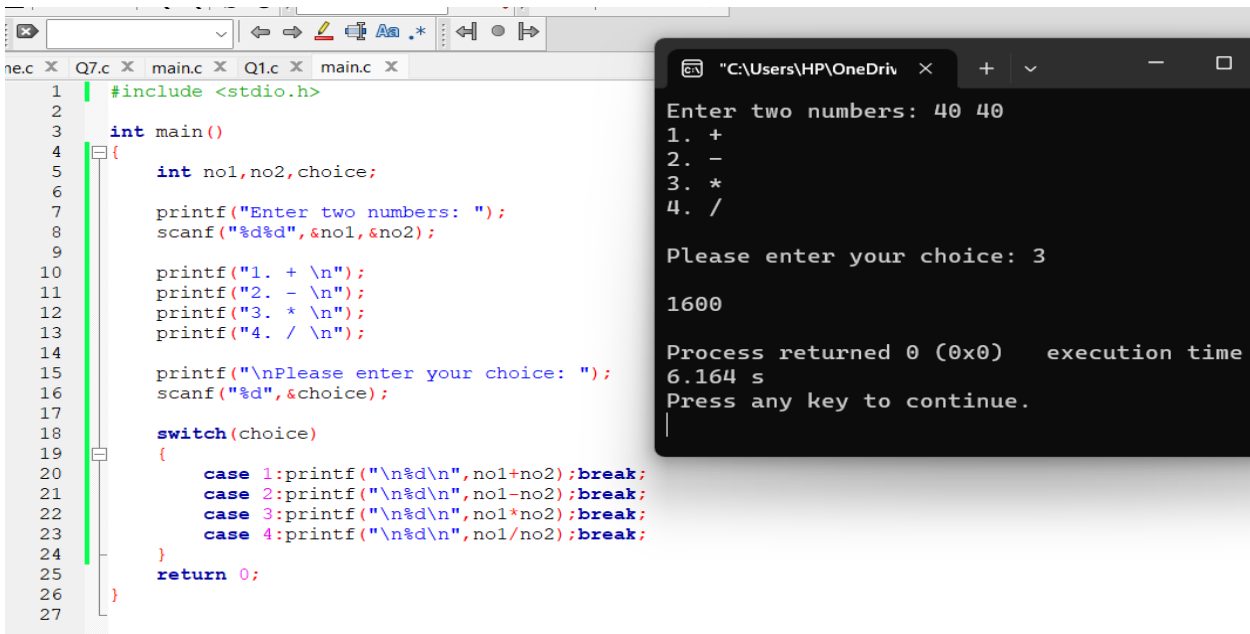
Switch

Input two numbers and display the outputs of the basic mathematic operations. The output screen should be displayed as follows;

Enter two numbers ____

1. +
2. -
3. *
4. /

Please enter your Choice ____



```
1 | #include <stdio.h>
2 |
3 | int main()
4 | {
5 |     int no1, no2, choice;
6 |
7 |     printf("Enter two numbers: ");
8 |     scanf("%d%d", &no1, &no2);
9 |
10 |    printf("1. + \n");
11 |    printf("2. - \n");
12 |    printf("3. * \n");
13 |    printf("4. / \n");
14 |
15 |    printf("\nPlease enter your choice: ");
16 |    scanf("%d", &choice);
17 |
18 |    switch(choice)
19 |    {
20 |        case 1: printf("\n%d\n", no1+no2); break;
21 |        case 2: printf("\n%d\n", no1-no2); break;
22 |        case 3: printf("\n%d\n", no1*no2); break;
23 |        case 4: printf("\n%d\n", no1/no2); break;
24 |    }
25 |    return 0;
26 | }
27 |
```

```
"C:\Users\HP\OneDrive" x + - □
Enter two numbers: 40 40
1. +
2. -
3. *
4. /

Please enter your choice: 3

1600

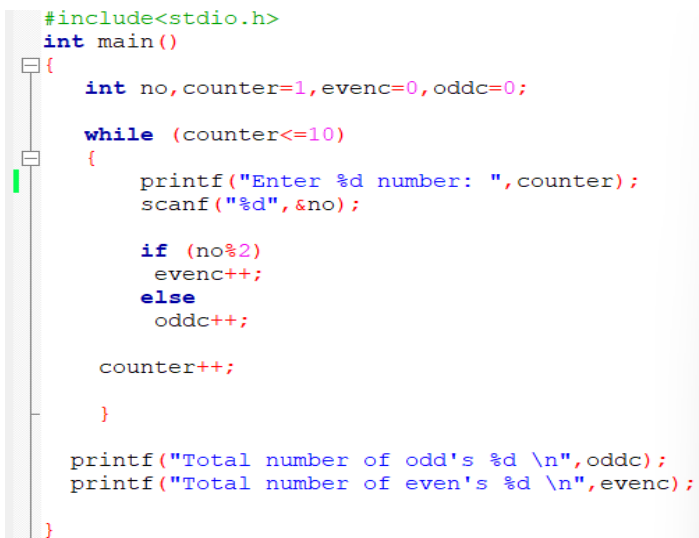
Process returned 0 (0x0)   execution time
6.164 s
Press any key to continue.
```

While loop

1. **Input 10 numbers and display the total count of odd & even numbers in the entered number series.**

```
#include<stdio.h>

int main()
{
    int no,counter=1,evenc=0,oddc=0;
    while (counter<=10)
    {
        printf("Enter %d number: ",counter);
        scanf("%d",&no);
        if (no%2)
            evenc++;
        else
            oddc++;
        counter++;
    }
    printf("Total number of odd's %d \n",oddc);
    printf("Total number of even's %d \n",evenc);
}
```



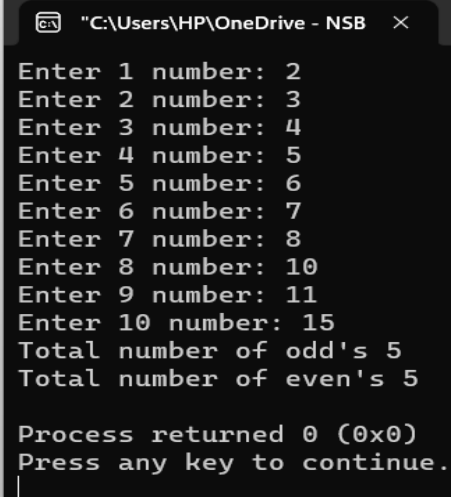
```
#include<stdio.h>
int main()
{
    int no,counter=1,evenc=0,oddc=0;

    while (counter<=10)
    {
        printf("Enter %d number: ",counter);
        scanf("%d",&no);

        if (no%2)
            evenc++;
        else
            oddc++;

        counter++;
    }

    printf("Total number of odd's %d \n",oddc);
    printf("Total number of even's %d \n",evenc);
}
```



```
"C:\Users\HP\OneDrive - NSB"
Enter 1 number: 2
Enter 2 number: 3
Enter 3 number: 4
Enter 4 number: 5
Enter 5 number: 6
Enter 6 number: 7
Enter 7 number: 8
Enter 8 number: 10
Enter 9 number: 11
Enter 10 number: 15
Total number of odd's 5
Total number of even's 5

Process returned 0 (0x0)
Press any key to continue.
|
```

2.Modify the above program in to enter series of numbers terminates when the user enter -99 and display the same expected output.

```
#include<stdio.h>
int main()
{
int no,evenc=0,oddc=0;
printf("Enter numbers (-99 to terminate)\n");
while (1)
{
printf("Number: ");
scanf("%d", &no);
if (no == -99)
{
break;
}
if (no%2)
evenc++;
else
oddc++;
}
printf("Total number of odd's %d \n",oddc);
printf("Total number of even's %d \n",evenc);

}
```

```

x ooo.c x while2.c x
#include<stdio.h>
int main()
{
    int no,evenc=0,oddc=0;

    printf("Enter numbers (-99 to terminate)\n");

    while (1)
    {
        printf("Number: ");
        scanf("%d", &no);

        if (no == -99)
        {
            break;
        }

        if (no%2)
            evenc++;
        else
            oddc++;
    }

    printf("Total number of odd's %d \n",oddc);
    printf("Total number of even's %d \n",evenc);
}

```

Enter numbers (-99 to terminate)

Number: 1

Number: 2

Number: 3

Number: 4

Number: 5

Number: 6

Number: 7

Number: 8

Number: 9

Number: 0

Number: 10

Number: 11

Number: 23

Number: 45

Number: -99

Total number of odd's 6

Total number of even's 8

Process returned 0 (0x0) execution time: 18.189 s

Press any key to continue.

Do while loop

Rewrite the programs for the above while loop question 1 & 2 using do while loop

```
1 #include<stdio.h>
   int main()
   {
       int no,counter=1,oddc=0,evenc=0;
       do
       {
           printf("Enter %d number : ",counter);
           scanf("%d",&no);

           if(no%2==0)
               evenc=evenc+1; //evenc++;
           else
               oddc=oddc+1;//oddc++;

           counter++;

       }while(counter<=10);

       printf("Total number of even's %d\n\n",evenc);
       printf("Total number of odd's %d\n",oddc);

       return 0;
   }
```

The image shows a C program in a code editor and its execution output in a terminal window. The code is a program that counts even and odd numbers from 1 to 10. The terminal output shows the user entering numbers 1 through 10, followed by the program's output: 'Total number of even's 5' and 'Total number of odd's 5'. The process returned 0 (0x0) and executed for 677 seconds.

```
#include<stdio.h>
int main()
{
    int no, counter=1, oddc=0, evenc=0;
    do
    {
        printf("Enter %d number : ", counter);
        scanf("%d", &no);

        if(no%2==0)
            evenc=evenc+1; //evenc++;
        else
            oddc=oddc+1; //oddc++;

        counter++;
    }while(counter<=10);

    printf("Total number of even's %d\n\n", evenc);
    printf("Total number of odd's %d\n", oddc);

    return 0;
}
```

Enter 1 number : 1
Enter 2 number : 2
Enter 3 number : 3
Enter 4 number : 4
Enter 5 number : 5
Enter 6 number : 6
Enter 7 number : 7
Enter 8 number : 8
Enter 9 number : 9
Enter 10 number : 10
Total number of even's 5

Total number of odd's 5

Process returned 0 (0x0) execut
677 s

2.

```
#include<stdio.h>
int main()
{
    int no, oddc=0, evenc=0;

    printf("Enter numbers (-99 to terminate)\n");

    do
    {
        printf("Number:");
        scanf("%d", &no);

        if (no == -99)
        {
            break;
        }
    }
}
```

```

    }

    if(no%2==0)
        evenc=evenc+1; //evenc++;
    else
        oddc=oddc+1;//oddc++;

}while(1);

printf("Total number of even's %d\n\n",evenc);
printf("Total number of odd's %d\n",oddc);

return 0;
}

```

```

re x ooo.c x while2.c x dowhile2.c x
1  #include<stdio.h>
2  int main()
3  {
4      int no,oddc=0,evenc=0;
5
6      printf("Enter numbers (-99 to terminate)\n");
7
8      do
9      {
10         printf("Number:");
11         scanf("%d",&no);
12
13         if (no == -99)
14         {
15             break;
16         }
17
18         if(no%2==0)
19             evenc=evenc+1; //evenc++;
20         else
21             oddc=oddc+1;//oddc++;
22     }while(1);
23
24     printf("Total number of even's %d\n\n",evenc);
25     printf("Total number of odd's %d\n",oddc);
26
27     return 0;
28 }
29
30

```

```

"C:\Users\HP\OneDrive - N x + v
Enter numbers (-99 to terminat
Number:2
Number:3
Number:4
Number:5
Number:6
Number:7
Number:8
Number:9
Number:29
Number:30
Number:99
Number:100
Number:-99
Total number of even's 6
Total number of odd's 6
Process returned 0 (0x0) exe
348 s
Press any key to continue.
|

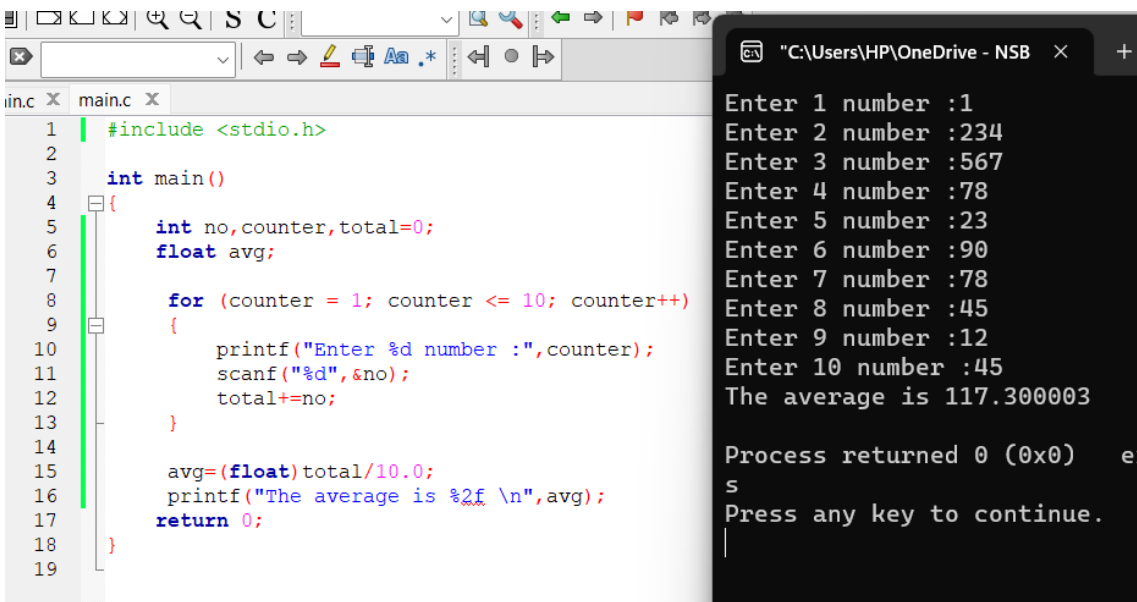
```

For loop

1. Input 10 numbers and display the average value using the for loop

```
#include <stdio.h>

int main()
{
    int no, counter, total=0;
    float avg;
    for (counter = 1; counter <= 10; counter++)
    {
        printf("Enter %d number :", counter);
        scanf("%d", &no);
        total+=no;
    }
    avg=(float)total/10.0;
    printf("The average is %2f \n", avg);
    return 0;
}
```



The screenshot shows a C program being executed in a terminal window. The program prompts the user to enter 10 numbers, which are then used to calculate the average. The output shows the average as 117.300003. The terminal window also displays the process return code and a prompt to press any key to continue.

```
inc X main.c X
1 | #include <stdio.h>
2 |
3 | int main()
4 | {
5 |     int no, counter, total=0;
6 |     float avg;
7 |
8 |     for (counter = 1; counter <= 10; counter++)
9 |     {
10 |         printf("Enter %d number :", counter);
11 |         scanf("%d", &no);
12 |         total+=no;
13 |     }
14 |
15 |     avg=(float)total/10.0;
16 |     printf("The average is %2f \n", avg);
17 |     return 0;
18 | }
19 |

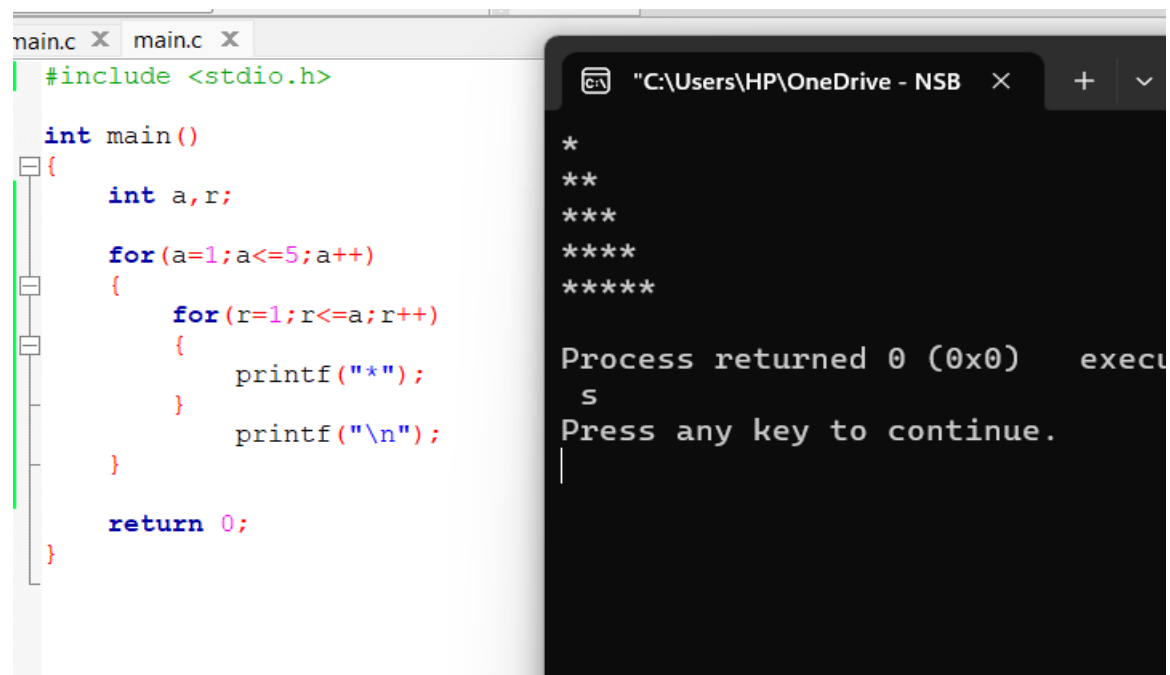
Enter 1 number :1
Enter 2 number :234
Enter 3 number :567
Enter 4 number :78
Enter 5 number :23
Enter 6 number :90
Enter 7 number :78
Enter 8 number :45
Enter 9 number :12
Enter 10 number :45
The average is 117.300003

Process returned 0 (0x0)   e
s
Press any key to continue.
|
```

2. Display the following output using the for loop

```
*  
**  
***  
****  
*****
```

```
#include <stdio.h>  
int main()  
{  
    int a,r;  
    for(a=1;a<=5;a++)  
    {  
        for(r=1;r<=a;r++)  
        {  
            printf("*");  
        }  
        printf("\n");  
    }  
    return 0;  
}
```



The screenshot shows a code editor window with the following C program:

```
main.c X main.c X  
#include <stdio.h>  
  
int main()  
{  
    int a,r;  
  
    for(a=1;a<=5;a++)  
    {  
        for(r=1;r<=a;r++)  
        {  
            printf("*");  
        }  
        printf("\n");  
    }  
  
    return 0;  
}
```

Next to the code editor is a terminal window showing the output of the program:

```
"C:\Users\HP\OneDrive - NSB X + v  
*  
**  
***  
****  
*****  
  
Process returned 0 (0x0)   execu  
s  
Press any key to continue.  
|
```